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Surviving winter hypoxia: behavioral adaptations of fishes in a northern Wisconsin winterkill lake

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Synopsis

Winterkill lakes often have a characteristic fish community, presumably composed of species able to survive winter hypoxia. Our research on a small winterkill lake in northern Wisconsin indicates that fishes common in winterkill lakes have behavioral adaptations for tolerating or avoiding winter hypoxia. We examined the distribution of fishes within the lake during one winter (December through May), and fish migrations into and out of the lake for two consecutive years. As DO within the lake declined in late fall, adult-sized fishes of four species, brook stickleback, finescale dace, redbelly dace, and fathead minnow, moved to the ice-water interface where DO levels were highest. Stickleback, and to a lesser extent, fathead minnows, also moved toward the more highly oxygenated water near the inlet. During the first year, young-of-the-year fishes of blacknose shiner, Iowa darter, redbelly dace, and fathead minnow, avoided hypoxic conditions by emigrating from the lake via the outlet stream in late fall and early winter while DO within the lake was still relatively high. Blacknose shiner, redbelly dace, and fathead minnow returned to the lake in spring. Almost no fishes were trapped leaving the lake in the second fall-winter season. Central mudminnows neither moved to the ice-water interface nor emigrated from the lake as DO dropped. Mudminnows survive winter hypoxia by breathing oxygen-containing bubbles trapped beneath the ice. These relatively simple behavioral adaptations allow fishes to survive or avoid hypoxic conditions lethal to other species and may help explain the consistency in fish communities of winterkill lakes.

Introduction

Many shallow, productive lakes in north temperate zones are severe environments for fishes in winter. Snow and ice normally obstruct light penetration, thus eliminating photosynthesis, and decomposing organic matter reduces oxygen concentrations (Greenbank 1945). The extent of oxygen depletion varies from lake to lake and from year to year within a lake. Lakes may become almost entirely anoxic, resulting in massive die-offs of some fish species (Greenbank 1945). Increased levels of ammonia, hydrogen sulfide, and carbon dioxide also occur at these times, but are believed to be of secondary importance to fish kills because dissolved oxygen (DO) declines to lethal levels before the other dissolved gases become harmful (Scidmore 1957).

Some fishes usually survive in winterkill lakes despite the severe conditions. Tonn & Magnuson (1982) found that northern Wisconsin lakes with a history of winterkill often have a characteristic fish community, presumably composed of species able to survive these severe, but variable conditions. In

laboratory experiments on three species common in winterkill lakes (central mudminnow, *Umbra limi;* fathead minnow, *Pimephales promelas;* brook stickleback, *Culaea inconstans*), Klinger et al. (1982) investigated the fishes' physiological adaptations to hypoxic conditions. Small size, low metabolic rates, and reduced activity under low DO conditions reduced the fishes' demands for oxygen; and head shape, increases in ventilation rates, and utilization of high oxygen microzones enhanced exploitation of available oxygen (Klinger et al. 1982). In this paper, we examine the behavioral adaptations of fishes that allow them to survive low oxygen conditions in a small winterkill lake in northern Wisconsin.

We hypothesized that as winter oxygen depletion proceeded some fishes would move to areas within the lake where levels of DO are highest, i.e., the ice-water interface or near the inlet where oxygenated water flows into the lake. Oxygen depletion generally begins at the bottom of the lake and proceeds upward during the winter so that the highest levels of oxygen occur at the ice-water interface (Greenbank 1945, Magnuson & Karlen 1970). Additionally, 0° C water entering the lake through a small inlet floats above the denser, warmer water, and forms a shallow layer of oxygenated water immediately beneath the ice (Mills 1972). We also hypothesized, that some fishes would emigrate from the lake into the more highly oxygenated inlet and outlet streams or to an adjacent deeper lake.

Methods

Our study was conducted in Mystery Lake, a 7 ha bog lake in Vilas County in northern Wisconsin (pH = 6.5, conductivity = 30 μmhs at 25° C, Black et al. 1963). This lake has a maximum depth of 1.6 m. Ice forms in late November and thaws in mid-April. Mystery Lake has a history of winterkill (about once in five years) with low oxygen levels ($<2\,mg\,l^{-1}$) usually occurring from late December through early March (Magnuson & Karlen 1970). Abundant fish species in the lake include blacknose shiner (*Notropis heterolepis*), fathead minnow, northern redbelly dace (*Phoxinus eos*), Iowa darter (*Etheostoma exile*), brook stickleback, central mudminnow, finescale dace (*Phoxinus neogaeus*), and yellow perch (*Perca flavescens*).

To investigate fish distributions during the winter we set wire minnow traps at 2-week intervals from 20 December 1970 through 4 May 1971, along a transect of seven pairs of stations extending from the inlet to the opposite shore (Fig. 1). Traps were 75 cm long and 45 cm in diameter with 0.64 mm meshes. Each station consisted of two trap sites 10 m apart. At each site two minnow traps were lowered through holes drilled in the ice. We placed one of the two traps on the bottom and suspended the other immediately below the ice. Owing to the thickness of the ice, only lower traps could be set at stations 1,2, and 7 from 4 February through 14 April 1971. Traps were set for approximately 48 h, but were emptied and baited with one half piece of

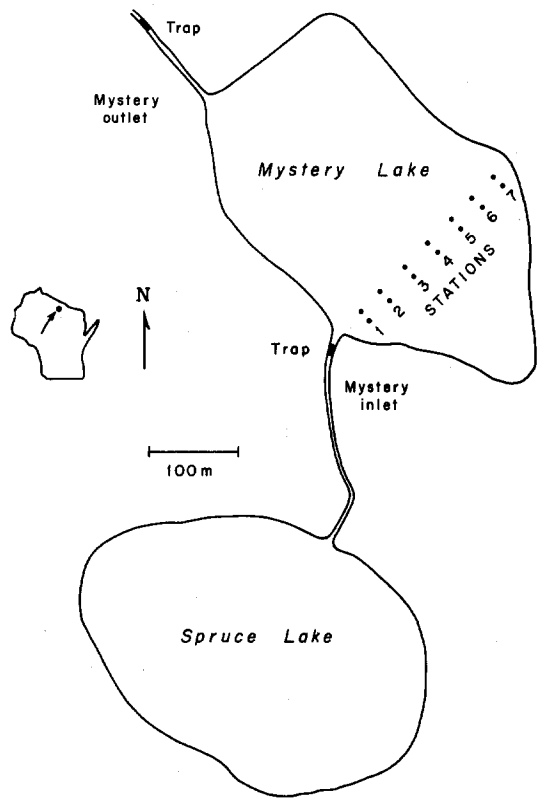

Fig. 1. Mystery and Spruce Lakes, Vilas County, Wisconsin. A transect of seven pairs of trapping stations extended from the Mystery Lake inlet to the opposite shore. Two-way traps were placed in the inlet and outlet of Mystery Lake.

white bread every 24 h. We identified, counted, and measured total length (to the nearest mm) of all fish caught. At each station we measured ice depth, trap depth, water temperature, dissolved oxygen and dissolved carbon dioxide. All water samples were fixed in the field and titrations were completed in the laboratory. DO was measured by the azide modification of the Winkler method (APHA 1965), and total free carbon dioxide, by the titration method (Golterman 1969). We measured water and air temperatures with mercury thermometers. Differences in oxygen and carbon dioxide levels and in water temperatures between stations near (stations 3 and 4) and away (stations 5 and 6) from the inlet and between upper and lower traps (stations 3 through 6), were tested with paired t-tests (Snedecor & Cochran 1967).

The Wilcoxon signed rank test (Snedecor & Cochran 1967) was used to test for differences in the vertical distribution of each species between upper and lower traps (stations 3 through 6) during each sampling period. To determine whether fishes aggregated near the inlet, we used Chi-square goodness-of-fit tests (Snedecor & Cochran 1967) to compare the number of each species caught at stations on the half of the sampling transect near the inlet (stations 3 and 4) with the number caught at stations on the half of the transect away from the inlet (stations 5 and 6) for each sampling period.

To determine whether drilling holes in the ice affected DO or trap catches, we evaluated DO and trap catches in holes and at locations under the ice 10 m from the holes at two different trapping stations. During eight sampling periods between 28 December 1971 and 5 March 1972, we measured DO and checked trap contents at the hole and 10 m from the hole 24 h and 48 h after the holes had been drilled. There were no significant differences in DO or in numbers of fish caught between the holes and locations under the ice (matched paired t-tests, $P>0.05$). It should be noted, however, that very few fish were caught within the lake in winter 1972.

Fish migrations into and out of Mystery Lake through the inlet and outlet were monitored by setting two-way traps in the streams. The inlet, a small, shallow stream, usually less than one meter deep and two meters wide, flows into Mystery Lake from Spruce Lake, a non-winterkill, 6.2 ha bog lake with a maximum depth of 4.9 m (pH = 6.2, conductivity = 16 μmhs, secchi = 2.6 m, Black et al. 1963). The outlet of Mystery Lake appears to disappear in a conifer swamp between Mystery Lake and Escanaba Lake, a 115 ha lake which does not winterkill. Reaeration of the outlet occurs as the water flows over a beaver dam and also in the areas below the dam.

Wooden traps with removable screens were placed permanently in the inlet and outlet (Fig. 1). The traps spanned completely the width and depth of the streams so that all fish traveling into or out of either stream had to pass through the traps. A screen partition in each trap separated fishes moving upstream from those moving downstream. We removed the screens between sampling periods. Trapping was conducted for ~48 h at biweekly intervals from October 1970 to September 1971 and from October 1971 to August 1972. The inlet was frozen solid from 23 January to 31 March 1972, so trapping at the inlet was suspended during that time. We emptied the traps each morning and afternoon during a sampling period, and all fish were identified, counted, and measured for total length. We took water and air temperatures and water samples for DO and CO_2 determinations once in the inlet and outlet during each sampling period.

Results

Vertical distributions within Mystery Lake

Five species, brook stickleback, redbelly dace, central mudminnow, finescale dace, and fathead minnow, were caught in sufficient numbers to examine within lake horizontal and vertical distributions. A few other species, blacknose shiner, silvery minnow (*Hybognathus nuchalis*), and Iowa darter, were caught in very low numbers within Mystery Lake.

Throughout the winter, fishes were found predominantly at the ice-water interface rather than the lake bottom. In all sampling periods from mid-December through late March (first of April for stickleback), significantly larger numbers of brook

stickleback, fathead minnow, redbelly dace, and finescale dace were caught in upper traps than in lower ones (P's <0.05, Table 1). In April there were no significant differences in the number of redbelly dace and finescale dace caught in upper and lower traps. Central mudminnows were distributed evenly during most of the winter and spring (Table 1). Nearly all fishes trapped within the lake were adult-sized, mean total length for stickleback was 51.3 mm, fathead minnow 61.0 mm, redbelly dace 64.2 mm, finescale dace 75.0 mm, and mudminnow 76.1 mm.

Chemical and physical conditions at the ice-water interface differed from those at the lake bottom throughout the winter and into early spring. In all sampling periods higher oxygen levels and lower temperatures and carbon dioxide levels occurred near the ice-water interface than near the bottom (matched t for oxygen = 5.7, for temperature = −8.94, for carbon dioxide = −5.01, P's <0.01) (Table 1).

Horizontal distributions within Mystery Lake

The brook stickleback was the only species that consistently aggregated near the inlet throughout the winter (Table 2). In December, when DO away from the inlet was still relatively high (4 mg l^{-1}), stickleback were distributed away from the inlet. During the five sampling periods from January through March, when DO away from the inlet was <2.2 mg l^{-1}, more stickleback were trapped near the inlet than away from the inlet (P's <0.05). From mid-January through the beginning of April, when oxygen levels away from the inlet were lowest, fathead minnows were more abundant near the inlet in three of the six sampling periods, and were never significantly more abundant away from the inlet (Table 2). Mudminnow and finescale dace distributions were unrelated to DO differences across the lake.

Conditions within Mystery Lake varied horizontally throughout the winter and into early spring.

Table 1. Vertical distribution of fishes in Mystery Lake during winter and early spring. The paired numbers given indicate the number of fish caught in upper and in lower traps. An arrow indicates that the difference between the numbers caught in upper traps and lower traps is significant (P<0.05). NS indicates no significant difference and * indicates that not enough fish were caught to test for differences. The final column is the percentage of sampling periods that fish were distributed toward the surface (includes only those sampling periods where in-lake DO≤4 mg l^{-1}).

Species	Dec 20–23	Jan 17–19	Feb 4–6	Feb 19–21	Mar 5–7	Mar 19–21	Apr 2–4	Apr 14–16	Sampling periods (%) fish were toward surface
Brook stickleback	↑ 199 144	↑ 599 94	↑ 500 30	↑ 719 86	↑ 391 53	↑ 749 167	↑ 151 64	↓ 215 619	100
Fathead minnows	↑ 73 12	↑ 75 19	↑ 45 1	↑ 66 4	↑ 72 5	↑ 15 4	* 9 0	* 0 6	100
Redbelly dace	↑ 56 14	↑ 159 15	↑ 78 0	↑ 324 11	↑ 146 5	↑ 73 13	NS 16 9	* 0 2	86
Finescale dace	↑ 87 5	↑ 73 18	↑ 39 2	↑ 71 19	↑ 40 10	↑ 27 14	NS 12 14	* 0 7	86
Mudminnows	NS 22 37	↓ 43 71	↑ 83 54	NS 45 57	NS 60 48	NS 23 38	NS 40 34	↓ 3 12	14
DO (mg l^{-1}) top	3.8	3.3	0.4	1.0	1.3	2.6	3.1	8.8	
DO (mg l^{-1}) bottom	1.0	0.4	0.0	0.1	0.1	0.1	0.5	6.6	
CO_2 (mg l^{-1}) top	14.7	26.6	10.2	17.4	34.0	21.9	17.8	10.1	
CO_2 (mg l^{-1}) bottom	15.6	32.0	14.8	25.1	38.8	26.7	18.3	15.6	
T (°C) top	1.7	0.8	0.7	0.3	0.0	0.5	0.3	2.0	
T (°C) bottom	3.6	2.7	1.5	2.1	1.0	1.8	1.7	3.0	

Oxygen levels were higher and water temperatures were lower near the inlet than away from the inlet (paired t for oxygen = 2.55, paired t for temperatures = −2.89, df = 7, P's <0.05) (Table 2). Carbon dioxide levels did not differ near and away from the inlet (paired t = −1.01, df = 7, P's <0.05). From December through April, the pH in Mystery Lake was 6.5 (range 6.5 to 6.6), mean pH within the inlet was 5.9 (range 5.2 to 6.6).

Emigration and immigration

Seven species, redbelly dace, fathead minnow, blacknose shiner, Iowa darter, central mudminnow, brook stickleback, and yellow perch, were caught in sufficient numbers in stream traps to assess movements into and out of Mystery Lake. Three types of migrations were observed: (1) emigration from the lake in late fall and early winter followed by immigration in spring (blacknose shiner, redbelly dace, fathead minnow, Iowa darter); (2) movements into and out of the lake during spring, presumably related to spawning (central mudminnow, brook stickleback, and yellow perch); and (3) dispersal of young-of-the-year in mid-summer (blacknose shiner and yellow perch). Because this research focused on behavioral adaptations for surviving winterkill conditions, only the first migration pattern will be discussed.

Fall-winter emigrations. – The 1970–71 fall-winter emigrations of blacknose shiner, redbelly dace, fathead minnow, and Iowa darter were remarkably synchronous. All four species had begun to leave Mystery Lake via the outlet in late November by the first sampling date with ice cover (Fig. 2). With the exception of blacknose shiner, the majority of fishes left the lake before DO had fallen to what we believe to be stressful levels (below 4 mg l^{-1}). Emigrations of all four species ended in mid-January when DO in the lake had reached its lowest level (Fig. 2).

Table 2. Horizontal distribution of fishes in Mystery Lake during winter and early spring. The number of fish caught in traps near (stations 3 and 4) and away from (stations 5 and 6) the inlet are given under the arrow which points to the left or right to indicate whether significantly more fish were caught near the inlet or away from the inlet (P<0.05). NS indicates no significant difference and * indicates that not enough fish were caught to test for differences. The final column is the percentage of sampling periods that fish were distributed toward the inlet (includes only those sampling periods where in-lake DO≤4 mg l^{-1}). DO, CO_2, and temperature were measured immediately below the ice, and values from stations 3, 4, 5, and 6 were used in the analyses.

Species	Dec 20–23	Jan 17–19	Feb 4–6	Feb 19–21	Mar 5–7	Mar 19–21	Apr 2–4	Apr 14–16	Sampling periods (%) fish were toward inlet
Brook stickleback	I →	I ←	I ←	I ←	I ←	I ←	I →	NS	71
	103 240	506 197	341 189	518 287	322 122	686 230	49 166	403 431	
Fathead minnows	NS	I ←	I ←	NS	I ←	NS	*	*	50
	49 36	61 33	42 4	33 37	65 12	10 9	0 9	2 4	
Redbelly dace	I ←	I →	I ←	I →	I ←	I →	I →	*	43
	61 9	70 104	59 19	177 218	88 63	18 68	0 25	0 2	
Finescale dace	I ←	NS	NS	I →	NS	NS	I →	*	14
	62 30	39 52	20 21	29 61	27 23	19 22	6 20	0 7	
Mudminnows	NS	NS	NS	NS	I →	NS	I ←	NS	14
	23 36	53 61	74 63	42 60	41 67	23 38	46 28	11 4	
DO (mg l^{-1}) toward inlet	3.6	5.5	0.6	1.4	2.1	3.6	4.0	8.9	
DO (mg l^{-1}) away from inlet	4.0	1.2	0.2	0.7	0.5	1.6	2.2	8.7	
CO_2 (mg l^{-1}) toward inlet	14.3	24.2	9.6	16.2	36.2	21.6	17.6	9.8	
CO_2 (mg l^{-1}) away from inlet	15.2	31.4	10.8	18.6	31.9	22.3	18.0	10.3	
T (°C) toward inlet	1.5	0.7	0.6	0.4	0.0	0.3	0.2	1.7	
T (°C) away from inlet	1.9	0.9	0.7	0.3	0.0	0.7	0.5	2.3	

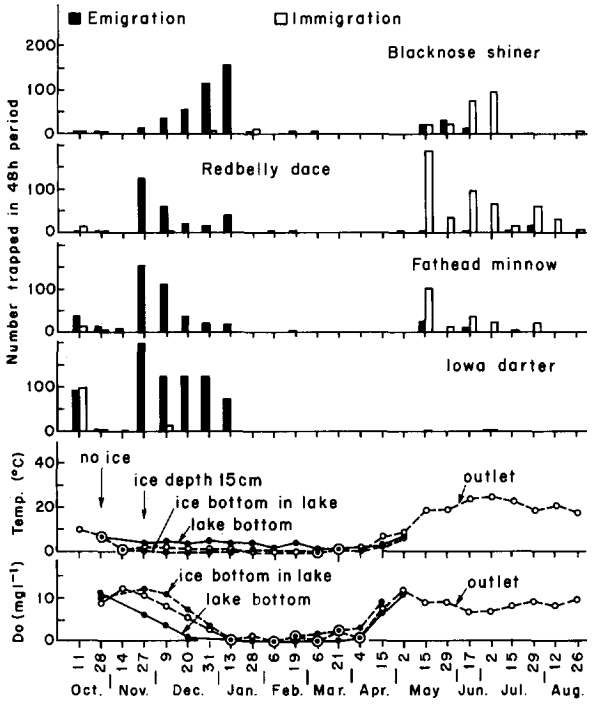

Fig. 2. Emigration and immigration via the outlet of blacknose shiner, redbelly dace, fathead minnow, and Iowa darter relative to temperatures and DO levels within Mystery Lake and in the outlet during 1970–71. Bars represent the number of fish caught during each 2-day trapping period.

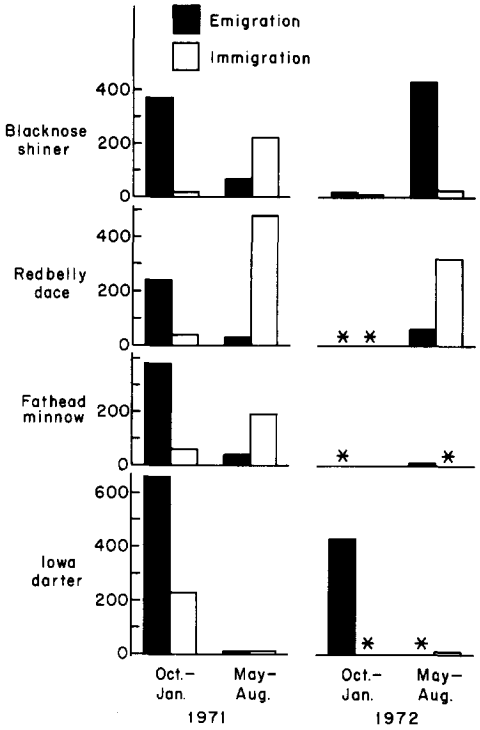

Fig. 3. Emigration and immigration of blacknose shiner, redbelly dace, fathead minnow, and Iowa darter via the outlet of Mystery Lake in late fall – early winter (October through January) and late spring – summer (May through August) in 1970–71 and 1971–72. Bars represent totals caught in biweekly sampling (* indicates <5 fish caught).

Large numbers of Iowa darter emigrated from Mystery Lake via the outlet in both 1970–71 and 1971–72 (Fig. 3). Fall-winter emigrations of blacknose shiner, redbelly dace, and fathead minnow were not evident the second year.

Fishes emigrating from the lake in fall and winter of 1970–71 were small in size and were probably young-of-the-year. Mean total length of emigrating blacknose shiner was 29.1 mm, redbelly dace 29.3 mm, fathead minnow 31.6 mm, and Iowa darter 39.5 mm (1970–71) and 46.9 mm (1971–72). Fishes remaining in Mystery Lake in winter were adult-sized. Mean total lengths of blacknose shiner (only 19 individuals), redbelly dace, and fathead minnow were 59.4 mm, 64.2 mm, and 61.0 mm, respectively (Fig. 4). Very few Iowa darter were trapped within the lake in winter.

No species emigrated from the lake via the inlet even though winter oxygen levels in the inlet remained high during the first winter ($\bar{x} = 8.7\,mg\,l^{-1}$, range $7.8–9.9\,mg\,l^{-1}$). During the second winter emigration via the inlet was impossible from 23 January to 31 March because the stream had frozen solid.

Spring-summer immigrations. – The 1971 spring-summer immigrations of blacknose shiner, redbelly dace, and fathead minnow were similar. Large numbers of adult-sized blacknose shiner, redbelly dace, and fathead minnow (mean total lengths = 42.0 mm, 52.7 mm, and 50.7 mm, respectively) began immigrating to Mystery Lake via the outlet in mid-May, four weeks after DO in the lake had begun to rise (Fig. 2). We believe many of these immigrants were fishes that had emigrated from the lake during the previous fall. They were intermediate in length between those emigrating in the fall and those adults remaining in the lake. The emigration of small blacknose shiner (mean total length = 28.7 mm) via the outlet in July of the

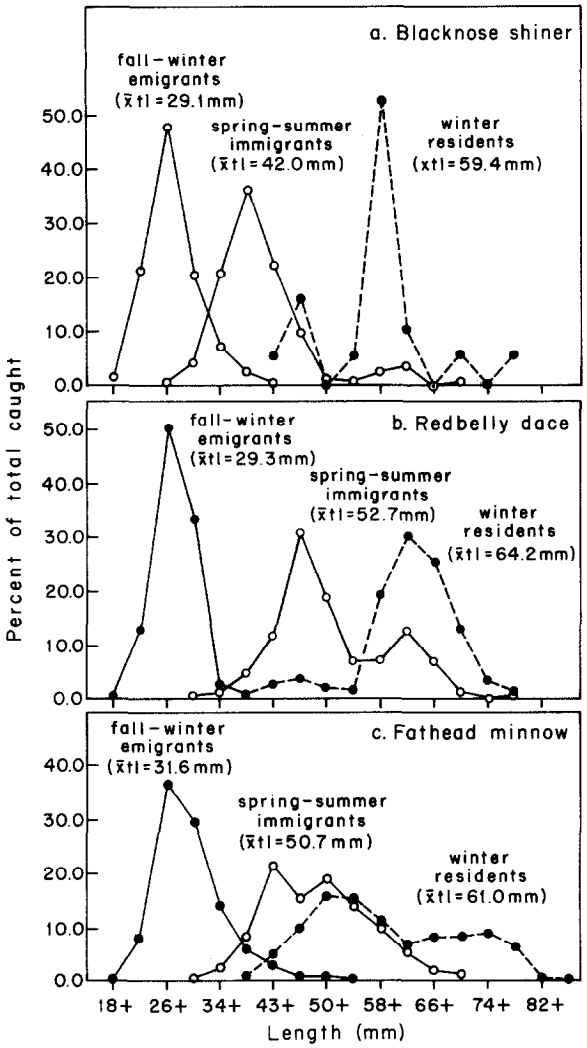

Fig. 4. Size distributions (percent composition) of blacknose shiner, redbelly dace, and fathead minnow emigrating from Mystery Lake (October through January), immigrating into the lake (May through August), and remaining in Mystery Lake (December through May) in 1970–71. x̄tl = mean total length.

second year (Fig. 3) probably represents dispersal of young-of-the-year.

Few Iowa darter were trapped returning to Mystery Lake in the spring of either year, a result possibly of the type of trap used, which may have allowed darter to swim under the trap, the periodicity of our sampling, or the life history of the fish. Iowa darter are negatively buoyant (Brown 1957). Swimming upstream, they may have been able to swim under the traps, whereas darter emigrating downstream could drift with the current and therefore, become caught in the traps. It is also possible that immigrations were missed by our biweekly sampling. Alternatively, a few fish may remain in the lake and spawn in the spring, with most young-of-the-year dispersing from the lake in the fall.

Discussion

Our data indicate that fishes may avoid the lethal effects of hypoxia in winterkill lakes by (1) moving to areas within the lake where levels of DO are higher, e.g. moving upward in the water column and/or horizontally toward the inlet or (2) emigrating from the lake into refugia, such as inlet and outlet streams. Different species used different strategies to cope with declining DO. The adults of two species, stickleback and finescale dace, moved to the ice-water interface; stickleback also aggregated at the inlet. Young-of-the-year blacknose shiner and Iowa darter emigrated from the lake into the more highly oxygenated outlet stream. Fathead minnow and redbelly dace exhibited a mixed strategy. Adults remained in the lake and moved to the ice-water interface as DO declined; young-of-the-year left the lake.

Very few adult-sized blacknose shiner or Iowa darter or young-of-the-year brook stickleback or finescale dace were trapped either within the lake or migrating into or out of Mystery Lake during our study. It is likely that these fishes were present, but possibly inactive during the winter. Similarly, although there was a large emigration of young-of-the-year yellow perch (1143, mean total length = 43.3 mm) from Mystery Lake in July and August 1971, few perch of any size were trapped leaving the lake or within Mystery Lake during the winter. How or whether these fishes tolerate winter hypoxia is unknown. We did not find large numbers of dead fish in the spring of either year of our study.

The finding that young-of-the-year fathead minnow and redbelly dace emigrate in fall, whereas adults of these species remain within the lake and move to the ice-water interface as DO declines indicates that young fathead minnows and redbelly dace may be less tolerant of low DO than are

adults. These results are consistent with the finding that large fish are more resistant to low oxygen tensions than are smaller fish of the same species at winter temperatures (Moore 1942, Shepard 1955), possibly owing to the generally higher metabolic rates of smaller fishes (Doudoroff & Shumway 1970). Casselman & Harvey (1975) found, however, that northern pike (*Esox lucius*) that died during partial winterkills were significantly larger than those that survived, indicating that smaller fish were more resistant to hypoxia.

Our results showing that some fishes move to the ice-water interface as levels of DO fall concur with laboratory experiments simulating winterkill conditions (Magnuson, unpublished data). Magnuson found that as DO declined from 4 mg l^{-1} to 0.5 mg l^{-1} stickleback increased the amount of time spent in the upper one-third of aquaria from 48 to 97 percent, fathead minnow from 53 to 69 percent, redbelly dace from 18 to 40 percent, and finescale dace from 28 to 44 percent.

Moving toward the surface appears to be a common response to declining levels of DO and may be a relatively simple, reflexive response. Magnuson & Karlen (1970) and Petrosky & Magnuson (1973) observed northern pike and yellow perch move upward as DO dropped in both laboratory and winter field experiments; these two species survived longer at low oxygen levels than did bluegills (*Lepomis macrochirus*) which did not move up toward the ice as DO declined. Similarly, a number of Great Plains fishes, including species in the families Catostomidae, Centrarchidae, Cyprinidae, Esocidae, and Percidae, moved toward the surface and irrigated their gills with the more highly oxygenated surface film when oxygen levels in aquaria dropped (Gee et al. 1978).

The stickleback was the only species that consistently aggregated near the inlet when levels of DO within the lake were low. Unlike stickleback, many cyprinids are intolerant of low pH (Rahel & Magnuson 1983, Magnuson et al. 1984) and may not have been attracted to the inlet because of the relatively low pH there, which ranged from 5.2 to 6.5 ($\bar{x} = 5.9$) during winter. The movement of stickleback toward the oxygenated water near the inlet indicates that these fish may be able to detect an oxygen gradient. Klinger et al. 1982 observed stickleback orienting to and irrigating their gills using the water from the opercular clefts of a mudminnow pumping oxygenated water down through a simulated ice crack in an aquarium.

The fall-winter emigrations of blacknose shiner, redbelly dace, fathead minnow, and Iowa darter were nearly synchronous and began when DO in the lake was still very abundant (>10 mg l^{-1}), especially at the ice-water interface. These findings suggest that the fishes' movements were prompted by a cue such as freeze over of the lake or the darkness following first snow on the ice rather than by oxygen stress. Winter oxygen depletion in small northern lakes is quite predictable. Species commonly found in these lakes may have evolved a behavior pattern of leaving these lakes long before DO declines to stressful levels. Cooper & Washburn (1949) and Johnson & Moyle (1969) reported that fishes moved into inlet and outlet streams as DO within lakes fell. And Tonn & Magnuson (1982) found that large fishes, such as northern pike, were present in shallow lakes with an inlet or outlet, but were present only in summer and not in winter, suggesting that in winter, these species emigrate from the lakes into stream refuges.

The lack of large fall emigrations of blacknose shiner, redbelly dace, and fathead minnow in 1971–72 is puzzling and cannot be explained by differences in levels of DO between years. The 1970–71 measurements of oxygen within the lake and at the outlet were similar to those made the previous year (paired t-test for DO in the lake at ice-water interface = -1.25, df = 3, $P>0.05$; for DO in outlet, t = -0.63, df = 13, $P>0.05$). During the second year all fish species were caught in substantially fewer numbers in all types of sampling gear. We hypothesize that (1) there may have been reproductive failure in spring 1971 for blacknose shiner, redbelly dace, and fathead minnow, (2) fish densities were reduced by our sampling methods, or (3) emigrations were missed by our biweekly sampling. Substantial mortality of fishes may have been caused by our sampling methods. Stream traps were operating about 14 percent of the time, so as much as 14 percent of fishes emigrating from the lake in winter, and 14 percent of those immigrating in spring

could have been caught. Trapping within the lake during winter 1970–71 caused additional mortality.

All fall emigrations occurred via the outlet rather than the inlet. The inlet may be unsuitable as a reliable emigration route and refuge because it has a relatively low pH, occasionally freezes solid, and originates from a lake containing large piscivores such as largemouth bass (*Micropterus salmoides*) (Tonn & Magnuson 1982). In winter the pH of the inlet ranged from 5.2 to 6.5 ($\bar{x}$ = 5.9), whereas outlet pH ranged from 6.0 to 6.6 ($\bar{x}$ = 6.4). Also, during the second year of our study the inlet was frozen solid from January 23 to March 31.

Central mudminnows, which survive well in winterkill lakes, neither moved to the ice-water interface nor emigrated from the lake as DO dropped. It is apparent that they must have other adaptations for surviving oxygen stress. Mudminnows are facultative air-breathers (Gee 1980), and Klinger et al. (1982) and Magnuson et al. (1983) found that mudminnows can survive hypoxic conditions in winter by breathing at oxygen-containing bubbles trapped beneath the ice.

Winterkill lakes are severe environments for most species. Because conditions in these lakes are apparently too stressful for many large piscivores, winterkill lakes may provide a refuge for those prey species that have evolved behavioral and physiological adaptations for surviving low oxygen conditions (Tonn & Magnuson 1982). Fathead minnow, redbelly dace, finescale dace, and stickleback simply moved to the ice-water interface as DO fell. Stickleback also moved toward the inlet. Blacknose shiner, Iowa darter, and young-of-the-year fathead minnow and redbelly dace avoided winterkill conditions by moving into the outlet. The ability of central mudminnows to breathe bubbles trapped beneath the ice, permit these fish to tolerate essentially anoxic water. These relatively simple behavioral and physiological adaptations allow these fishes to survive conditions lethal to other species and may help to explain the consistency observed in fish communities among winterkill lakes (Tonn & Magnuson 1982).

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